

25PY101: Engineering Physics

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References

Module 2 pre-T1 slot2

1 Topics

Topics	Avadhanulu	Neamen
E_F in ext. s.c.		
	37.15 Ext s.c.	4.2 Dopant atoms and energy levels
	37.16 n-type	4.3.1 Equilibrium electron and hole concentration
	37.17 p-type	
Band diagrams of n-type and p-type		
	37.18 Band diagrams @ $T = 0\text{ K}$ and $T \neq 0\text{ K}$	4.4.2 Complete ionization and freeze out
	37.23 E_F in ext. s.c.	4.6 Position of Fermi level
pn junction diode – Forward and reverse bias		
	38.2 pn junction diode	7.1 Basic structure of pn junction
	38.3 pn junction diode under forward bias	7.2 Zero applied bias
	38.4 pn junction diode under reverse bias	7.3 Reverse applied bias
	38.6 $I - V$ characteristic	8.1.1 Qualitative description of charge in pn junction
		8.1.5 Ideal pn junction current
Solar cell		
	38.12 Solar cell	14.1 Optical absorption
		14.2.1 pn junction solar cell
		14.2.2 Conversion efficiency and solar concentration
Direct and indirect band gap semiconductors		
	37.27 Compound semiconductors	3.3.1 k-space diagram of Si and Ge

2 Test your understanding

E_F in extrinsic semiconductor

1. What is a donor impurity? What is an acceptor impurity?
2. Differentiate between n-type and p-type semiconductor by sketching their band diagrams.
3. p-type and n-type have more holes and electrons respectively. Explain why they are electrically neutral.

4. What is meant by complete ionization? What is meant by freeze-out?
5. What is the effect of temperature on the working of n-type and p-type semiconductors? Draw energy band diagrams to show the variation in the position of Fermi level with rise in temperature.
6. Draw energy band diagrams for n-type semiconductor at 0 K and at room temperature.
7. Distinguish between extrinsic and intrinsic semiconductors.
8. Sketch a graph of n_0 versus temperature for an n-type semiconductor.
9. Sketch a graph of E_F versus temperature for p-type semiconductor.
10. What is the relevance of the Fermi energy?

pn junction diode

1. What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]
2. Explain the formation of **potential barrier** across the pn junction region.
3. Explain the formation of **depletion region** in the pn junction region. What is the typical value of depletion region width for commonly used pn junctions?
4. Draw band diagrams for symmetrically doped pn junction at
 - zero bias
 - forward bias
 - reverse bias
5. What is reverse saturation current in a diode?
6. Why is reverse saturation current independent of reverse bias?
7. What is reverse breakdown?

Solar cell

1. Explain the working of solar cell.
2. Draw and explain the $I - V$ characteristics of a solar cell.
3. A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?
4. An alien ship crashlanded onto Earth’s surface. From the debris, the solar panel used for power generation is found to have a band gap of 5 eV. Estimate the temperature of the alien Sun.

Direct and indirect band gap semiconductors

1. Compare and contrast elemental and compound semiconductors.
2. Define direct and indirect band gap semiconductor.
3. Describe optical absorption process in semiconductors. When is optical absorption essentially zero?
4. (i) What are the particles involved during the optical absorption in a direct band gap semiconductor. Write the reaction between the particles. (ii) Repeat part (i) for the indirect band gap semiconductor.
5. Why are compound semiconductors useful for optoelectronic applications?
6. Is silicon a direct or indirect band gap semiconductor? Which type of semiconductor is useful for optoelectronic applications? Do you see a dilemma? Discuss how silicon is used in solar cell applications.

3 Technical terms

E_F in extrinsic semiconductor

- acceptor atoms
- donor atoms
- extrinsic semiconductor
- complete ionization
- freeze-out

pn junction diode

- metallurgical junction
- diffusion
- built-in electric field
- thermal equilibrium
- built-in potential barrier
- depletion region
- space charge region
- space charge width
- reverse saturation current
- reverse breakdown

Solar cell

- optical absorption
- generation of excess charge carriers
- conversion efficiency
- fill factor
- open-circuit voltage
- short-circuit current

Direct and indirect band gap semiconductors

- Elemental semiconductor
- Compound semiconductor
- Direct band gap semiconductor
- Indirect band gap semiconductor
- Radiative recombination of charge carriers